

```
import pandas as pd
import numpy as np

df = pd.read_excel('/content/icc_dataset (2).xlsx')

df.head()
```

	Tournament	Year	Venue	Winner	Runner-up	Semi-Finalists	Match_Winner_Final
0	ODI World Cup	1975	England	West Indies	Australia	WI, AUS, NZ, ENG	Clive Lloyd
1	ODI World Cup	1979	England	West Indies	England	WI, ENG, PAK, NZ	Viv Richards
2	ODI World Cup	1983	England	India	West Indies	IND, WI, PAK, ENG	M. Amarnath

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
# Problem 1: Find total number of data entries using NumPy
num_entries = df.to_numpy().size
print(f"Total number of data entries: {num_entries}")
```

Total number of data entries: 248

```
# Problem 2: Find unique tournament types using NumPy
unique_tournaments = np.unique(df['Tournament'].to_numpy())
print(f"Unique tournament types: {unique_tournaments}")
```

Unique tournament types: ['Champions Trophy' 'ODI World Cup' 'T20 World Cup']

```
# Problem 3: Find earliest year using NumPy
earliest_year = np.min(df['Year'].to_numpy())
print(f"Earliest year: {earliest_year}")
```

Earliest year: 1975

```
# Problem 4: Find latest year using NumPy
latest_year = np.max(df['Year'].to_numpy())
print(f"Latest year: {latest_year}")
```

Latest year: 2025

```
# Problem 5: Find number of unique venues using NumPy
num_unique_venues = len(np.unique(df['Venue'].to_numpy()))
```

```
print(f"Number of unique venues: {num_unique_venues}")
```

Number of unique venues: 15

```
# Problem 6: Convert winner column into NumPy array
winner_array = df['Winner'].to_numpy()
print(f"Winner column as NumPy array: {winner_array[:5]}...") # Display first 5
```

Winner column as NumPy array: ['West Indies' 'West Indies' 'India' 'Australia' 'South Africa']

```
# Problem 7: Count total number of winners using NumPy
total_winners = winner_array.size
print(f"Total number of winners: {total_winners}")
```

Total number of winners: 31

```
# Problem 8: Find unique winners using NumPy
unique_winners = np.unique(winner_array)
print(f"Unique winners: {unique_winners}")
```

Unique winners: ['Australia' 'England' 'IND/SL Joint' 'India' 'New Zealand' 'Pakistan' 'South Africa' 'Sri Lanka' 'West Indies']

```
# Problem 9: Find common teams in winner and runner-up using NumPy
runner_up_array = df['Runner-up'].to_numpy()
common_teams = np.intersect1d(unique_winners, np.unique(runner_up_array))
print(f"Common teams in winner and runner-up: {common_teams}")
```

Common teams in winner and runner-up: ['Australia' 'England' 'India' 'New Zealand' 'Sri Lanka' 'West Indies']

```
# Problem 10: Count number of unique years using NumPy
unique_years_count = len(np.unique(df['Year'].to_numpy()))
print(f"Number of unique years: {unique_years_count}")
```

Number of unique years: 29

✓ Solving problems using Pandas

```
# Problem 1: Find total number of data entries using Pandas
num_entries_pd = df.size
print(f"Total number of data entries (Pandas): {num_entries_pd}")
```

Total number of data entries (Pandas): 248

```
# Problem 2: Find unique tournament types using Pandas
unique_tournaments_pd = df['Tournament'].unique()
print(f"Unique tournament types (Pandas): {unique_tournaments_pd}")
```

Unique tournament types (Pandas): ['ODI World Cup' 'T20 World Cup' 'Champions Tr

```
# Problem 3: Find earliest year using Pandas
earliest_year_pd = df['Year'].min()
print(f"Earliest year (Pandas): {earliest_year_pd}")
```

Earliest year (Pandas): 1975

```
# Problem 4: Find latest year using Pandas
latest_year_pd = df['Year'].max()
print(f"Latest year (Pandas): {latest_year_pd}")
```

Latest year (Pandas): 2025

```
# Problem 5: Find number of unique venues using Pandas
num_unique_venues_pd = df['Venue'].nunique()
print(f"Number of unique venues (Pandas): {num_unique_venues_pd}")
```

Number of unique venues (Pandas): 15

```
# Problem 6: Convert winner column into Pandas Series
winner_series = df['Winner']
print(f"Winner column as Pandas Series (first 5 entries):\n{winner_series.head(
```

Winner column as Pandas Series (first 5 entries):

0 West Indies

1 West Indies

2 India

3 Australia

4 Pakistan

Name: Winner, dtype: object

```
# Problem 7: Count total number of winners using Pandas
total_winners_pd = winner_series.size
print(f"Total number of winners (Pandas): {total_winners_pd}")
```

Total number of winners (Pandas): 31

```
# Problem 8: Find unique winners using Pandas
unique_winners_pd = winner_series.unique()
print(f"Unique winners (Pandas): {unique_winners_pd}")
```

Unique winners (Pandas): ['West Indies' 'India' 'Australia' 'Pakistan' 'Sri Lank
'South Africa' 'New Zealand' 'IND/SL Joint']

```
# Problem 9: Find common teams in winner and runner-up using Pandas
runner_up_series = df['Runner-up']
common_teams_pd = pd.Series(list(set(unique_winners_pd).intersection(set(runner
print(f"Common teams in winner and runner-up (Pandas): {common_teams_pd.to_numpy
```

```
Common teams in winner and runner-up (Pandas): ['Pakistan' 'West Indies' 'South
'Sri Lanka' 'New Zealand']
```

```
# Problem 10: Count number of unique years using Pandas
unique_years_count_pd = df['Year'].nunique()
print(f"Number of unique years (Pandas): {unique_years_count_pd}")
```

```
Number of unique years (Pandas): 29
```